Oral 1 Solutions

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Problem 3

We call such a sequence $a_1, \ldots, a_n \in \mathbb{Z}^d$ π-coverable for a permutation π of $1, \ldots, d$, if the following algorithm returns `true`:

Algorithm 1: Let $k := 1$, $s_1, \ldots, s_n := \infty$. For $i := 1, \ldots, n$, if there exists $1 \leq j < k$ s.t. $s_{\pi_j} = a_{i,\pi_j}$, then we continue; otherwise, if $k > d$, we return `false`, otherwise we set $s_{\pi_k} := a_{i,\pi_k}$ and increment $k := k+1$. The algorithm returns `true` if the outer loop terminates. Note that this algorithm is straightforward to generalize to any partial permutation ρ.

Lemma 1: A sequence $a_1, \ldots, a_n$ is coverable iff it is π-coverable for some $\pi \in$ permutations$(1, \ldots, d)$.
Proof:

- ($\leftarrow$): Let π be some permutation for which the algorithm returns `true`. Consider the terminating values of k and $s_1, \ldots, s_d$ for such a π. It is clear that for each $i = 1, \ldots, n$, there exists some $1 \leq j < k$ s.t. $s_x = a_{i,x}$ for $x = p_j$. Thus, s is a covering sequence of a.
- ($\rightarrow$): Let t be a covering sequence of a. We construct a π s.t. a is π-coverable by repeating the following process: let i be the index of the first uncovered point; since t covers a_i, there exists $1 \leq j \leq d$ s.t. $t_j = a_{i,j}$. We append such a j to π, and mark every point ℓ s.t. $t_j = a_{\ell,j}$ as covered. This is the exact reverse process as the forward algorithm, so it yields π s.t. a is π-coverable.

Let $f_{i,\rho}$ for $1 \leq i \leq n$ and ρ a partial permutation of $1, \ldots, d$ be the maximum index $i \leq j \leq n$ s.t. the subsequence $a_i, \ldots, a_j$ is ρ-coverable. The base case of the recurrence is given by $f_{i,\varnothing} = i - 1$ for all $1 \leq i \leq n$ (corresponding to a degenerate interval).

Suppose that $|\rho| = m$. Let j^* be the minimum index in $1, \ldots, m$ s.t.

$$a_{\left(f_{i+1,\overline{\rho_2 \cdots \rho_{j^*}}}\right)+1,\rho_1} = a_{i,\rho_1}.$$

If such a j^* exists, then let $\psi = \overline{\rho_2 \cdots \rho_{j^*} \rho_1 \rho_{j^*+1} \cdots \rho_m}$. We have that $f_{i,\rho} = f_{i+1,\psi}$. Otherwise, $f_{i,\rho} = f_{i+1,\overline{\rho_2 \cdots \rho_m}}$. Evaluating in reverse order from n to 1, we have a recurrence for f.

To motivate this solution, we first note that $f_{i,\rho}$ is equivalent to running Algorithm 1 with fixed permutation ρ from index i in a. Thus, we must have that $s_{\rho_1} = a_{i,\rho_1}$. The formula above finds some partial permutation ψ s.t. Algorithm 1, when starting from index $i+1$, will generate the same solution. We do this by identifying the minimum prefix of $\overline{\rho_2 \cdots \rho_m}$; if the first index that cannot be covered (index $f_{i+1,\overline{\rho_2 \cdots \rho_j}} + 1$) has ρ_1-th index equivalent to the ρ_1-th index at a_1, then for the current index, it can be matched at dimension ρ_1; hence, this is equivalent to $f_{i+1,\psi}$, as defined above. If no such index exists, then index ρ_1 is never "useful" outside of index a_i, so the answer is $f_{i+1,\overline{\rho_2 \cdots \rho_m}}$.

Then, the answer is given by

$$\sum_{i=1}^{n} \left(\max_{\pi \in \text{permutations}(1,\ldots,d)} f_{i,\pi} \right) - i + 1.$$

To show that this is correct, we introduce the following lemma.

Lemma 2: If a sequence $a_1, \ldots, a_n$ is coverable, then the sequence $a_1, \ldots, a_{n-1}$ is coverable.
Proof: Trivial.

Thus, for each position i, finding the maximum $i \leq j \leq n$ s.t. the subinterval from i to j is coverable necessarily shows the existence of $j - i + 1$ subintervals that have left boundary at i and are coverable. For such a maximal j, by Lemma 1 there exists some permutation π s.t. the subinterval is π-coverable; enumerating over all π is thus sufficient to identify the maximum.

There are $O(d!)$ partial permutations, yielding $O(nd!)$ states. The transitions have a complexity bounded by $O(d)$, giving a total complexity of $O(nd!d)$.